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SE Lab 7

Initial processor.py execution

```
PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): 0-306-40615-2
Group 306: Starts 0, Checksum 2

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/10
Depreciation Factor: 0.7

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): MINT
Multiplier: 1.25

PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): abc
Traceback (most recent call last):
  File "C:\bookstore\Lab7\processor.py", line 40, in <module>
    main()
  File "C:\bookstore\Lab7\processor.py", line 29, in main
    print(verify_isbn10_format(isbn_in))
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\bookstore\Lab7\processor.py", line 3, in verify_isbn10_format
    first_digit = int(cleaned[0])
                  ^^^^^^^^^^^^^^
ValueError: invalid literal for int() with base 10: 'a'
```

Failing Hypothesis test cases

```
PS C:\bookstore\Lab7> python -m pytest test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\bookstore\Lab7
plugins: anyio-4.14.2, hypothesis-6.170.14, platformdirs-4.13.0
collected 3 items

test_processor.py FFF                                                    [100%]

================================== FAILURES ===================================
_____________________ test_verify_isbn10_format_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\bookstore\Lab7\test_processor.py", line 5, in test_verify_isbn10_format_no_crash
  |     def test_verify_isbn10_format_no_crash(s):
  |                   ^^^
  |   File "C:\Users\Asus\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
```

```
|     result = verify_isbn10_format(s)
|              ^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\bookstore\Lab7\processor.py", line 3, in verify_isbn10_format
|     first_digit = int(cleaned[0])
|                   ^^^^^^^^^^^^^^^
| ValueError: invalid literal for int() with base 10: ':'
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
|     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
| AssertionError: verify_isbn10_format crashed on ':' with: ValueError: invalid literal for int() with base 10: ':'
| assert False
| Failing test case: test_verify_isbn10_format_no_crash(
|     s=':',
| )
| Explanation:
|     These lines were always and only run by failing test cases:
|         C:\bookstore\Lab7\test_processor.py:8
+---------------- 2 ----------------
| Traceback (most recent call last):
|   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
|     result = verify_isbn10_format(s)
|              ^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\bookstore\Lab7\processor.py", line 5, in verify_isbn10_format
|     check_char = cleaned[9].upper()
|                  ~~~~~~~~^^^
| IndexError: string index out of range
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
|     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
| AssertionError: verify_isbn10_format crashed on '0' with: IndexError: string index out of range
| assert False
| Failing test case: test_verify_isbn10_format_no_crash(
|     s='0',
| )
| Explanation:
|     These lines were always and only run by failing test cases:
|         C:\bookstore\Lab7\test_processor.py:8
+---------------- 3 ----------------
| Traceback (most recent call last):
|   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
|     result = verify_isbn10_format(s)
|              ^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\bookstore\Lab7\processor.py", line 3, in verify_isbn10_format
|     first_digit = int(cleaned[0])
|                       ~~~~~~~~^^^
| IndexError: string index out of range
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
|     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
| AssertionError: verify_isbn10_format crashed on '' with: IndexError: string index out of range
| assert False
| Failing test case: test_verify_isbn10_format_no_crash(
|     s='',
| )
| Explanation:
|     These lines were always and only run by failing test cases:
|         C:\bookstore\Lab7\test_processor.py:8
+---------------- 4 ----------------
| Traceback (most recent call last):
|   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
|     result = verify_isbn10_format(s)
|              ^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\bookstore\Lab7\processor.py", line 2, in verify_isbn10_format
|     cleaned = isbn_str.replace('-', '').strip()
|               ^^^^^^^^^^^^^^^^
| AttributeError: 'NoneType' object has no attribute 'replace'
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
|     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
| AssertionError: verify_isbn10_format crashed on None with: AttributeError: 'NoneType' object has no attribute 'replace'
| assert False
| Failing test case: test_verify_isbn10_format_no_crash(
|     s=None,  # or any other generated value
| )
```

```
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:8
    +------------------------------------
_____________________ test_calculate_depreciation_factor_safe _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\bookstore\Lab7\test_processor.py", line 13, in test_calculate_depreciation_factor_safe
  |     def test_calculate_depreciation_factor_safe(s):
  |                    ^^^
  |   File "C:\Users\Asus\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 15, in test_calculate_depreciation_factor_safe
    |     result = calculate_depreciation_factor(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 10, in calculate_depreciation_factor
    |     age = float(age_str)
    |           ^^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 17, in test_calculate_depreciation_factor_safe
    |     assert False, f"calculate_depreciation_factor crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: calculate_depreciation_factor crashed on '/' with: ValueError: could not convert string to float: ''
    | assert False
    | Failing test case: test_calculate_depreciation_factor_safe(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 15, in test_calculate_depreciation_factor_safe
    |     result = calculate_depreciation_factor(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 9, in calculate_depreciation_factor
    |     age_str, life_str = book_age_str.split('/')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 17, in test_calculate_depreciation_factor_safe
    |     assert False, f"calculate_depreciation_factor crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: calculate_depreciation_factor crashed on '' with: ValueError: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_calculate_depreciation_factor_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 15, in test_calculate_depreciation_factor_safe
    |     result = calculate_depreciation_factor(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 9, in calculate_depreciation_factor
    |     age_str, life_str = book_age_str.split('/')
    |                         ^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 17, in test_calculate_depreciation_factor_safe
    |     assert False, f"calculate_depreciation_factor crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: calculate_depreciation_factor crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_calculate_depreciation_factor_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_lookup_condition_multiplier_safe _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\bookstore\Lab7\test_processor.py", line 21, in test_lookup_condition_multiplier_safe
  |     def test_lookup_condition_multiplier_safe(s):
  |                    ^^^
  |   File "C:\Users\Asus\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 23, in test_lookup_condition_multiplier_safe
    |     result = lookup_condition_multiplier(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
    |   File "C:\bookstore\Lab7\processor.py", line 24, in lookup_condition_multiplier
    |     return multipliers[grade]
    |            ~~~~~~~~~~~~~~^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 25, in test_lookup_condition_multiplier_safe
    |     assert False, f"lookup_condition_multiplier crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: lookup_condition_multiplier crashed on '' with: KeyError: ''
    | assert False
    | Failing test case: test_lookup_condition_multiplier_safe(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:24
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 23, in test_lookup_condition_multiplier_safe
    |     result = lookup_condition_multiplier(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 16, in lookup_condition_multiplier
    |     grade = condition_code.strip().upper()
    |             ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 25, in test_lookup_condition_multiplier_safe
    |     assert False, f"lookup_condition_multiplier crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: lookup_condition_multiplier crashed on None with: AttributeError: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_lookup_condition_multiplier_safe(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:24
    +----------------------------------
========================== short test summary info ==========================
FAILED test_processor.py::test_verify_isbn10_format_no_crash - ExceptionGroup...
FAILED test_processor.py::test_calculate_depreciation_factor_safe - Exception...
FAILED test_processor.py::test_lookup_condition_multiplier_safe - ExceptionGr...
============================ 3 failed in 2.44s ============================
```

Fixed processor.py execution

```
PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): 0-306-40615-2
Group 306: Starts 0, Checksum 2

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/10
Depreciation Factor: 0.7

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): MINT
Multiplier: 1.25

PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): abc
Invalid ISBN-10 format

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/0
Depreciation Factor: Invalid age/lifespan

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): FAIR
Multiplier: Unknown condition

PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2):
Invalid ISBN-10 format

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): -2/10
Depreciation Factor: Invalid age/lifespan

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR):   good
Multiplier: 0.6
```

Successful Hypothesis test execution

```
PS C:\bookstore\Lab7> python -m pytest test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\bookstore\Lab7
plugins: anyio-4.14.2, hypothesis-6.170.14, platformdirs-4.13.0
collected 3 items

test_processor.py ...                                                    [100%]

============================== 3 passed in 1.83s ==============================
```